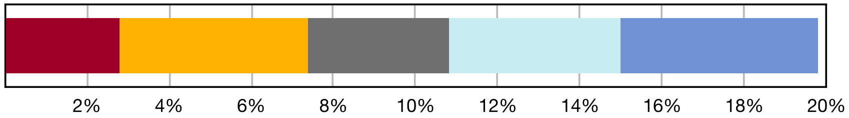
# ADVANCED
**Stage 6**

## Calculus (Adv), C3 Applications of Calculus (Adv)
### Curve Sketching (Y12)

**Teacher:** SHS
**Exam Equivalent Time:** 42 minutes *(based on allocation of 1.5 minutes per mark)*

## C3 Applications of Calculus




- The Derivative Function and its Graph
- Curve Sketching
- Tangents
- Maxima and Minima
- Rates of Change

***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

### HISTORICAL CONTRIBUTION

- *C3 Applications of Calculus* is the 800 pound gorilla in the Advanced rainforest, contributing a massive 19.8% to new syllabus exams since introduced in 2020.
- This topic has been split into five sub-topics for analysis purposes: *1-The Derivative Function and its Graph (2.8%), 2-Curve Sketching (4.6%), 3-Tangents (3.4%), 4-Maxima and Minima (4.2%) and 5-Rates of Change (4.8%).*
- This analysis looks at *Curve Sketching.*

### HSC ANALYSIS - What to expect and common pitfalls

- *Curve Sketching (4.6%)* is consistently examined with significant mark allocations.
- The most popular curve is easily the cubic, asked 6 times in the last decade, including each year in the period 2017-2020.
- Important: The *2022 Adv 22* cubic asked for the "global maximum and minimum" within a given range for the first time. This required finding the $y$-values of the graph at the domain limits - a must review question!
- Sketches of polynomials of degree 4 are the second most popular, most recently tested in 2024 (see *2024 Adv 19* and *2012 Adv 14a*).
- *Curve Sketching* questions regularly require students to look at concavity and intervals where a curve is increasing or decreasing. An important revision area that proiduces low mean marks.

## Questions

### 1. Calculus, 2ADV C3 2022 HSC 22

Find the global maximum and minimum values of $y = x^3 - 6x^2 + 8$, where $-1 \leq x \leq 7$. ***(4 marks)***

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### 2. Calculus, 2ADV C3 2004 HSC 9c

Consider the function $f(x) = \dfrac{\log_e x}{x}$, for $x > 0$.

i. Show that the graph of $y = f(x)$ has a stationary point at $x = e$. ***(2 marks)***

ii. By considering the gradient on either side of $x = e$, or otherwise, show that the stationary point is a maximum. ***(1 mark)***

iii. Use the fact that the maximum value of $f(x)$ occurs at $x = e$ to deduce that $e^x \geq x^e$ for all $x > 0$. ***(2 marks)***

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### 3. Calculus, 2ADV C3 2009 HSC 10

Let $f(x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3}$

a. Show that the graph of $y = f(x)$ has no turning points. ***(2 marks)***

b. Find the point of inflection of $y = f(x)$. ***(1 mark)***

c. **i.** Show that $1 - x + x^2 - \dfrac{1}{1+x} = \dfrac{x^3}{1+x}$ for $x \neq -1$. ***(1 mark)***

**ii.** Let $g(x) = \ln(1+x)$.
Use the result in part **c.i.** to show that $f'(x) \geq g'(x)$ for all $x \geq 0$. ***(2 marks)***

d. Sketch the graphs of $y = f(x)$ and $y = g(x)$ for $x \geq 0$. ***(2 marks)***

e. Show that $\dfrac{d}{dx}[(1+x)\ln(1+x) - (1+x)] = \ln(1+x)$. ***(2 marks)***

f. Find the area enclosed by the graphs of $y = f(x)$ and $y = g(x)$, and the straight line $x = 1$. ***(2 marks)***

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### 4. Calculus, 2ADV C3 2012 HSC 14a

A function is given by $f(x) = 3x^4 + 4x^3 - 12x^2$.

i. Find the coordinates of the stationary points of $f(x)$ and determine their nature. ***(3 marks)***

ii. Hence, sketch the graph $y = f(x)$ showing the stationary points. ***(2 marks)***

iii. For what values of $x$ is the function increasing? ***(1 mark)***

iv. For what values of $k$ will $f(x) = 3x^4 + 4x^3 - 12x^2 + k = 0$ have no solution? ***(1 mark)***

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## Worked Solutions



### 1. Calculus, 2ADV C3 2022 HSC 22

$$y = x^3 - 6x^2 + 8$$
$$\frac{dy}{dx} = 3x^2 - 12x$$
$$\frac{d^2y}{dx^2} = 6x - 12$$

SP's when $\dfrac{dy}{dx} = 0$:

$$3x^2 - 12x = 0$$
$$3x(x-4) = 0$$

$x = 0$ or $4$

When $x = 0,\ y = 8,\ \dfrac{d^2y}{dx^2} < 0$

$\Rightarrow$ Local Max at $(0, 8)$

When $x = 4,\ y = 4^3 - 6(4^2) + 8 = -24,\ \dfrac{d^2y}{dx^2} > 0$

$\Rightarrow$ Local Min at $(4, -24)$

Check ends of domain:
When $x = -1,\ y = -1 - 6 + 8 = 1$
When $x = 7,\ y = 7^3 - 6(7^2) + 8 = 57$
$\therefore$ Global max $= 57$
$\therefore$ Global min $= -24$

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### 2. Calculus, 2ADV C3 2004 HSC 9c

**i.** $f(x) = \dfrac{\log_e x}{x}$, for $x > 0$

Using the quotient rule,

$$f\prime\left(\frac{u}{v}\right) = \frac{u\prime v - uv\prime}{v^2}$$

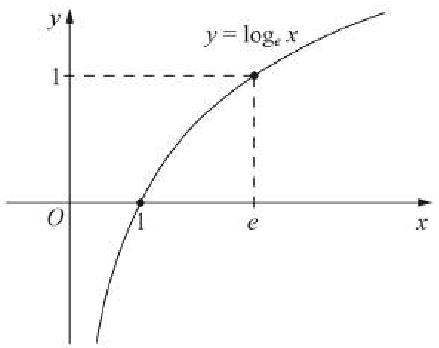
$$f\prime(x) = \frac{x \cdot \frac{d}{dx}(\log_e x) - \log_e x \cdot \frac{d}{dx}(x)}{x^2}$$
$$= \frac{x\left(\frac{1}{x}\right) - \log_e x \cdot 1}{x^2}$$
$$= \frac{1 - \log_e x}{x^2}$$

SP's occur when $f\prime(x) = 0$

$$\frac{1 - \log_e x}{x^2} = 0$$
$$1 - \log_e x = 0$$
$$\log_e x = 1$$
$$\therefore x = e$$

$\therefore$ There is a stationary point at $x = e$.

**ii.**




When $x < e, \log_e x < 1$ (from graph)

$$1 - \log_e x > 0$$
$$\frac{1 - \log_e x}{x^2} > 0$$
$$\therefore f\prime(x) > 0$$

When $x > e, \log_e x > 1$ (from graph)

$$1 - \log_e x < 0$$
$$< 0$$

$$\frac{1-\log_e x}{x^2}$$

$\therefore f\prime(x) < 0$

$\therefore$ The stationary point at $x = e$ is a maximum.

**iii.** $f(e) = \dfrac{\log_e x}{e}$

$= \dfrac{1}{e}$

$\therefore f(x)$ has a maximum value at $\left(e, \dfrac{1}{e}\right)$

**MARKER'S COMMENT:** Very challenging for most students. Successful students recognised the link to part (ii).

$$\begin{aligned}
\therefore \frac{\log_e x}{x} &\le \frac{1}{e} \\
\log_e x &\le \frac{x}{e} \\
e\log_e x &\le x \\
\log_e x^e &\le x \\
e^{\log_e x^e} &\le e^x \\
x^e &\le e^x \quad (\text{using } e^{\log x} = x) \\
\therefore e^x &\ge x^e \quad (\text{for } x > 0)
\end{aligned}$$

## 3. Calculus, 2ADV C3 2009 HSC 10

♦♦ *Mean mark 28% for all of Q10 (note that data for each question part is not available).*

**a.** $f(x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3}$

Turning points when $f\prime(x) = 0$

$f\prime(x) = 1 - x + x^2$

$x^2 - x + 1 = 0$

$$\begin{aligned}
\text{Since } \Delta &= b^2 - 4ac \\
&= (-1)^2 - 4 \times 1 \times 1 \\
&= -3 < 0 \Rightarrow \text{No solution}
\end{aligned}$$

$\therefore f(x)$ has no turning points

**b.** P.I. when $f\prime\prime(x) = 0$

$f\prime\prime(x) = -1 + 2x = 0$

$2x = 1$

$x = \dfrac{1}{2}$

Check for change in concavity

$f\prime\prime\left(\dfrac{1}{4}\right) = -\dfrac{1}{2} < 0$

$f\prime\prime\left(\dfrac{3}{4}\right) = \dfrac{1}{2} > 0$

$\Rightarrow$ Change in concavity

$\therefore$ P.I. at $x = \dfrac{1}{2}$

$$\begin{aligned}
f\left(\frac{1}{2}\right) &= \frac{1}{2} - \frac{\left(\frac{1}{2}\right)^2}{2} + \frac{\left(\frac{1}{2}\right)^3}{3} \\
&= \frac{1}{2} - \frac{1}{8} + \frac{1}{24} \\
&= \frac{5}{12}
\end{aligned}$$

$\therefore$ Point of Inflection at $\left(\dfrac{1}{2}, \dfrac{5}{12}\right)$

**c.i.** Show $1 - x + x^2 - \dfrac{1}{1+x} = \dfrac{x^3}{1+x}, \quad x \ne -1$

$$\begin{aligned}
\text{LHS} &= \frac{1+x}{1+x} - \frac{x(1+x)}{1+x} + \frac{x^2(1+x)}{(1+x)} - \frac{1}{1+x} \\
&= \frac{1 + x - x - x^2 + x^2 + x^3 - 1}{1+x}
\end{aligned}$$

$$= \frac{x^3}{1+x} \quad \ldots \text{ as required}$$

**c.ii.** Let $g(x) = \ln(1+x)$

$$g\prime(x) = \frac{1}{1+x}$$

$$f\prime(x) - g\prime(x) = 1 - x + x^2 - \frac{1}{1+x}$$

$$= \frac{x^3}{1+x} \quad \text{(using part (i))}$$

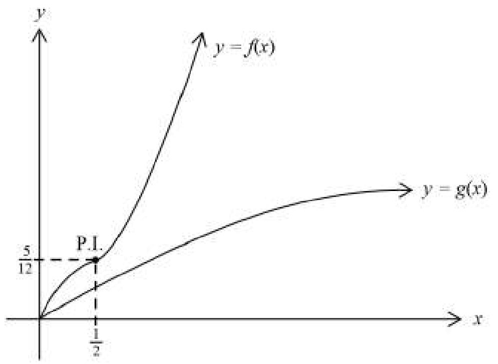
Since $\frac{x^3}{1+x} \geq 0$ for $x \geq 0$

$$f\prime(x) - g\prime(x) \geq 0$$

$$f\prime(x) \geq g\prime(x) \text{ for } x \geq 0$$

**MARKER'S COMMENT:** When 2 graphs are drawn on the same set of axes, you ***must*** label them.

**d.**




**e.** Show $\frac{d}{dx}[(1+x)\ln(1+x) - (1+x)] = \ln(1+x)$

Using $\frac{d}{dx}uv = uv\prime + vu\prime$

$$\text{LHS} = (1+x) \times \frac{1}{1+x} + \ln(1+x) \times 1 + -1$$

$$= 1 + \ln(1+x) - 1$$

$$= \ln(1+x)$$

$$= \text{RHS} \quad \ldots \text{ as required}$$

**f.** Area $= \int_0^1 f(x) - g(x)\ dx$

$$= \int_0^1 \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \ln(x+1)\right) dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{12} - (1+x)\ln(1+x) + (1+x)\right]_0^1$$

(using part (e) above)

$$= \left[\left(\frac{1}{2} - \frac{1}{6} + \frac{1}{12} - (2)\ln 2 + 2\right) - (\ln 1 + 1)\right]$$

$$= \frac{5}{12} - 2\ln 2 + 2 - 1$$

$$= 1\frac{5}{12} - 2\ln 2 \quad \text{u}^2$$

4. Calculus, 2ADV C3 2012 HSC 14a

**i.**

$$f(x) = 3x^4 + 4x^3 - 12x^2$$
$$f'(x) = 12x^3 + 12x^2 - 24x$$
$$f''(x) = 36x^2 + 24x - 24$$

Stationary points when $f'(x) = 0$

$$12x^3 + 12x^2 - 24x = 0$$
$$12x\left(x^2 + x - 2\right) = 0$$
$$12x(x+2)(x-1) = 0$$

$\therefore$ Stationary points at $x = 0,\ 1$ or $-2$

When $x = 0,\quad f(0) = 0$

$f''(0) = -24 < 0$

$\therefore$ MAX at (0,0)

When $x = 1$

$f(1) = 3 + 4 - 12 = -5$

$f''(1) = 36 + 24 - 24 = 36 > 0$

$\therefore$ MIN at $(1, -5)$

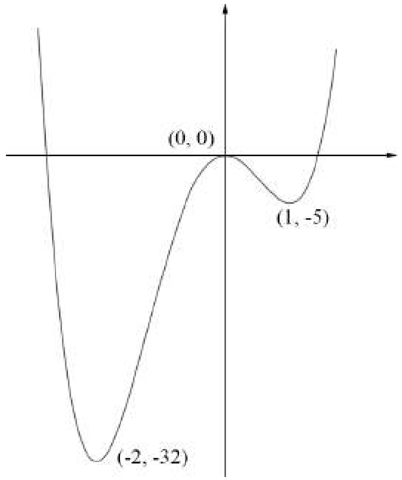
When $x = -2$

$$\begin{aligned} f(-2) &= 3(-2)^4 + 4(-2)^3 - 12(-2)^2 \\ &= 48 - 32 - 48 \\ &= -32 \end{aligned}$$

$$\begin{aligned} f''(-2) &= 36(-2)^2 + 24(-2) - 24 \\ &= 144 - 48 - 24 = 72 > 0 \end{aligned}$$

$\therefore$ MIN at (–2, –32)

**ii.**




**iii.** $f(x)$ is increasing for

$-2 < x < 0$ and $x > 1$

♦ *Mean mark 42%*
**MARKER'S COMMENT:** Be careful to use the correct inequality signs, and not carelessly include ≥ or ≤ by mistake.

**iv.** Find $k$ such that

$3x^4 + 4x^3 - 12x^2 + k = 0$ has no solution

$k$ is the vertical shift of $y = 3x^4 + 4x^3 - 12x^2$

$\Rightarrow$ No solution if it does not cross the $x$-axis.

$\therefore$ No solution when $k > 32$

♦♦♦ *Mean mark 12%.*